American Computer Science League

2020-2021 ● Contest 1: Shorts ● Senior Division

1. Computer Number Systems

How many 6's are there in the octal representation of the decimal numbers from 675 to 700, inclusive?

A. 9
B. 10
C. 11
D. 12
E. 13

2. Computer Number Systems

Evaluate and express in base 16 if both numbers are hexadecimal.

13 * 74F

A. $8BDD_{16}$
B. $8ADD_{16}$
C. $7BDE_{16}$
D. $8AEC_{16}$
E. $8ACD_{16}$

3. Recursive Functions

Find $f(24)$, given:

$$f(x) = \begin{cases} x + f(x-5) & \text{if } x \geq 10 \\ f(x+3) - 4 & \text{if } 5 < x < 10 \\ 2x - 1 & \text{if } x \leq 5 \end{cases}$$

A. 9
B. 49
C. 56
D. 76
E. 80

4. Recursive Functions

Find $f(3, 18)$, given:

$$f(x,y) = \begin{cases} f(x-2, y+3) + 3 & \text{if } x > y \\ f(y-1, x+1) - 5 & \text{if } x < y \\ xy & \text{if } x = y \end{cases}$$

A. 139
B. 142
C. 143
D. 144
E. 148

5. What Does This Program Do?

When this program is executed, what is the largest value of any variable?

```
a = 16: b = 4 : d =12
c = a / b
if a / c > d / b then
        if a < 15 then
                d = b + c
        else
                c = a + b
        end if
else
        if b > 3 then
                a = a  + d / b
        end if
end if
if c > a then
        t = a
        a = c
        c = t
end if
if a > b + c  || a > d && a > b then
        d = d + (a - c) ^ 2
else
        a = a + b * d
end if
```

A. 28
B. 20
C. 16
D. 12
E. 8